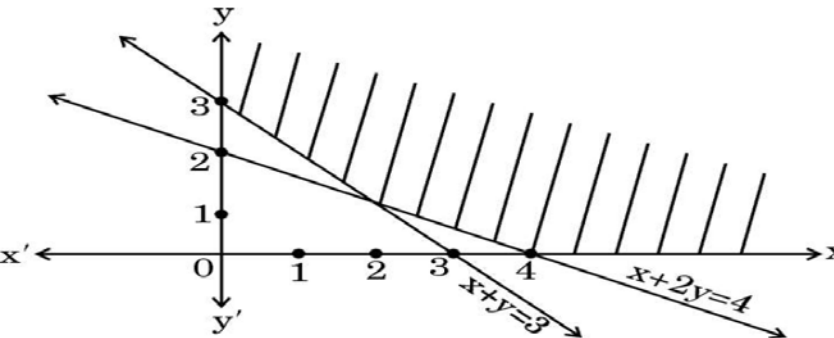


LINEAR PROGRAMMING

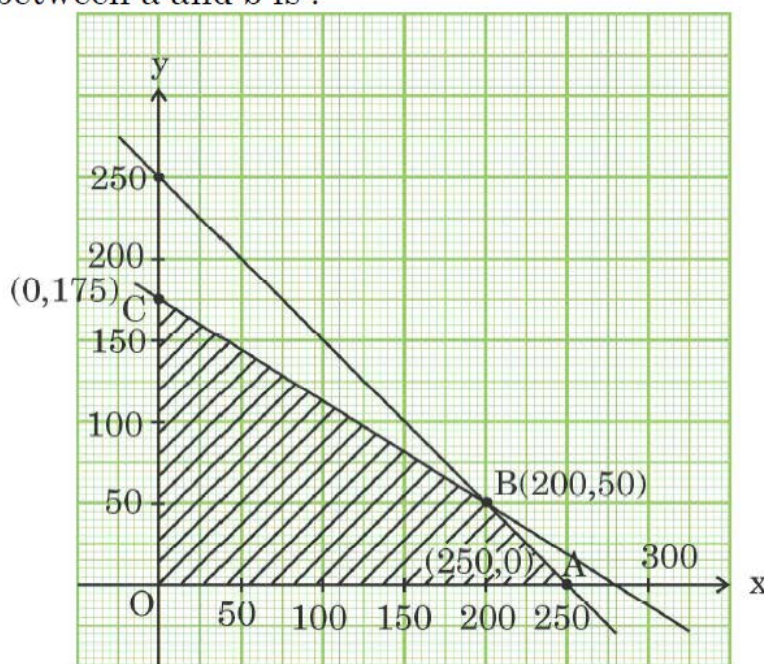
1.	The corner points of the feasible region in the graphical representation of a linear programming problem are $(2, 72)$, $(15, 20)$ and $(40, 15)$. If $z = 18x + 9y$ be the objective function, then : (a) z is maximum at $(2, 72)$, minimum at $(15, 20)$ (b) z is maximum at $(15, 20)$, minimum at $(40, 15)$ (c) z is maximum at $(40, 15)$, minimum at $(15, 20)$ (d) z is maximum at $(40, 15)$, minimum at $(2, 72)$
2.	The number of corner points of the feasible region determined by the constraints $x - y \geq 0$, $2y \leq x + 2$, $x \geq 0$, $y \geq 0$ is : (a) 2 (b) 3 (c) 4 (d) 5
3.	Solve graphically the following linear programming problem : Maximise $z = 6x + 3y$, subject to the constraints $4x + y \geq 80,$ $3x + 2y \leq 150,$ $x + 5y \geq 115,$ $x \geq 0, y \geq 0.$
4.	The objective function $Z = ax + by$ of an LPP has maximum value 42 at $(4, 6)$ and minimum value 19 at $(3, 2)$. Which of the following is true ? (A) $a = 9, b = 1$ (B) $a = 5, b = 2$ (C) $a = 3, b = 5$ (D) $a = 5, b = 3$
5.	The corner points of the feasible region of a linear programming problem are $(0, 4)$, $(8, 0)$ and $\left(\frac{20}{3}, \frac{4}{3}\right)$. If $Z = 30x + 24y$ is the objective function, then (maximum value of Z – minimum value of Z) is equal to (A) 40 (B) 96 (C) 120 (D) 136

6.	Solve the following linear programming problem graphically : Maximize : $Z = x + 2y$ subject to constraints : $x + 2y \geq 100$, $2x - y \leq 0$, $2x + y \leq 200$, $x \geq 0, y \geq 0$.
7.	The number of feasible solutions of the linear programming problem given as Maximize $z = 15x + 30y$ subject to constraints : $3x + y \leq 12, x + 2y \leq 10, x \geq 0, y \geq 0$ is (a) 1 (b) 2 (c) 3 (d) infinite
8.	The feasible region of a linear programming problem is shown in the figure below :  Which of the following are the possible constraints ? (a) $x + 2y \geq 4, x + y \leq 3, x \geq 0, y \geq 0$ (b) $x + 2y \leq 4, x + y \leq 3, x \geq 0, y \geq 0$ (c) $x + 2y \geq 4, x + y \geq 3, x \geq 0, y \geq 0$ (d) $x + 2y \geq 4, x + y \geq 3, x \leq 0, y \leq 0$
9.	Solve the following linear programming problem graphically : Minimize $z = x + 2y$ subject to the constraints $2x + y \geq 3$, $x + 2y \geq 6$, $x \geq 0$, $y \geq 0$.

10.	The solution set of the inequation $3x + 5y < 7$ is : (a) whole xy-plane except the points lying on the line $3x + 5y = 7$. (b) whole xy-plane along with the points lying on the line $3x + 5y = 7$. (c) open half plane containing the origin except the points of line $3x + 5y = 7$. (d) open half plane not containing the origin.
11.	Which of the following points satisfies both the inequations $2x + y \leq 10$ and $x + 2y \geq 8$? (a) $(-2, 4)$ (b) $(3, 2)$ (c) $(-5, 6)$ (d) $(4, 2)$
12.	Solve the following linear programming problem graphically : Minimise : $z = -3x + 4y$ subject to the constraints $x + 2y \leq 8,$ $3x + 2y \leq 12,$ $x, y \geq 0.$
13.	Solve the following Linear Programming Problem graphically : Maximize : $P = 70x + 40y$ subject to : $3x + 2y \leq 9,$ $3x + y \leq 9,$ $x \geq 0, y \geq 0$

14.	Solve the following linear programming problem graphically : Minimise $Z = 6x + 7y$ subject to constraints $x + 2y \geq 240$ $3x + 4y \leq 620$ $2x + y \geq 180$ $x, y \geq 0.$
15.	The point which lies in the half-plane $2x + y - 4 \leq 0$ is : (a) (0, 8) (b) (1, 1) (c) (5, 5) (d) (2, 2)
16.	Solve the following Linear Programming Problem graphically : Minimise $z = 3x + 8y$ subject to the constraints $3x + 4y \geq 8$ $5x + 2y \geq 11$ $x \geq 0, y \geq 0$

- 17.** The corner points of the bounded feasible region of an LPP are $O(0, 0)$, $A(250, 0)$, $B(200, 50)$ and $C(0, 175)$. If the maximum value of the objective function $Z = 2ax + by$ occurs at the points $A(250, 0)$ and $B(200, 50)$, then the relation between a and b is :



- (a) $2a = b$ (b) $2a = 3b$ (c) $a = b$ (d) $a = 2b$

- 18.** A linear programming problem deals with the optimization of a/an :

- (A) logarithmic function (B) linear function
(C) quadratic function (D) exponential function

- 19.** The number of corner points of the feasible region determined by constraints $x \geq 0$, $y \geq 0$, $x + y \geq 4$ is :

- (A) 0 (B) 1
(C) 2 (D) 3

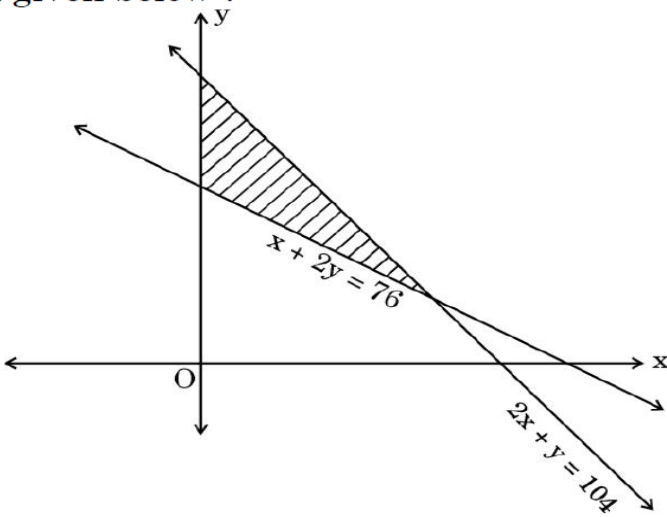
- 20.** Solve the following linear programming problem graphically :
Maximise $z = 500x + 300y$,
subject to constraints

$$x + 2y \leq 12$$

$$2x + y \leq 12$$

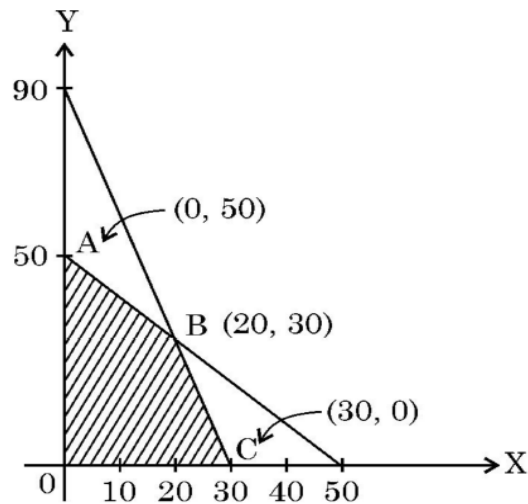
$$4x + 5y \geq 20$$

$$x \geq 0, y \geq 0$$

21.	The common region determined by all the constraints of a linear programming problem is called : (A) an unbounded region (B) an optimal region (C) a bounded region (D) a feasible region
22.	Of the following, which group of constraints represents the feasible region given below ?  (A) $x + 2y \leq 76$, $2x + y \geq 104$, $x, y \geq 0$ (B) $x + 2y \leq 76$, $2x + y \leq 104$, $x, y \geq 0$ (C) $x + 2y \geq 76$, $2x + y \leq 104$, $x, y \geq 0$ (D) $x + 2y \geq 76$, $2x + y \geq 104$, $x, y \geq 0$
23.	The restrictions imposed on decision variables involved in an objective function of a linear programming problem are called : (A) feasible solutions (B) constraints (C) optimal solutions (D) infeasible solutions
24.	Solve the following linear programming problem graphically : Maximize $z = x + y$ subject to constraints $2x + 5y \leq 100$ $8x + 5y \leq 200$ $x \geq 0, y \geq 0.$

25.

The maximum value of $Z = 4x + y$ for a L.P.P. whose feasible region is given below is :



(A) 50

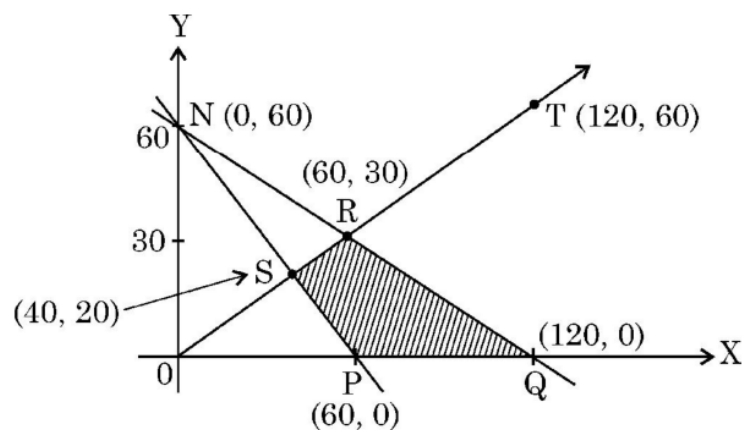
(B) 110

(C) 120

(D) 170

26.

Assertion (A) : The corner points of the bounded feasible region of a L.P.P. are shown below. The maximum value of $Z = x + 2y$ occurs at infinite points.

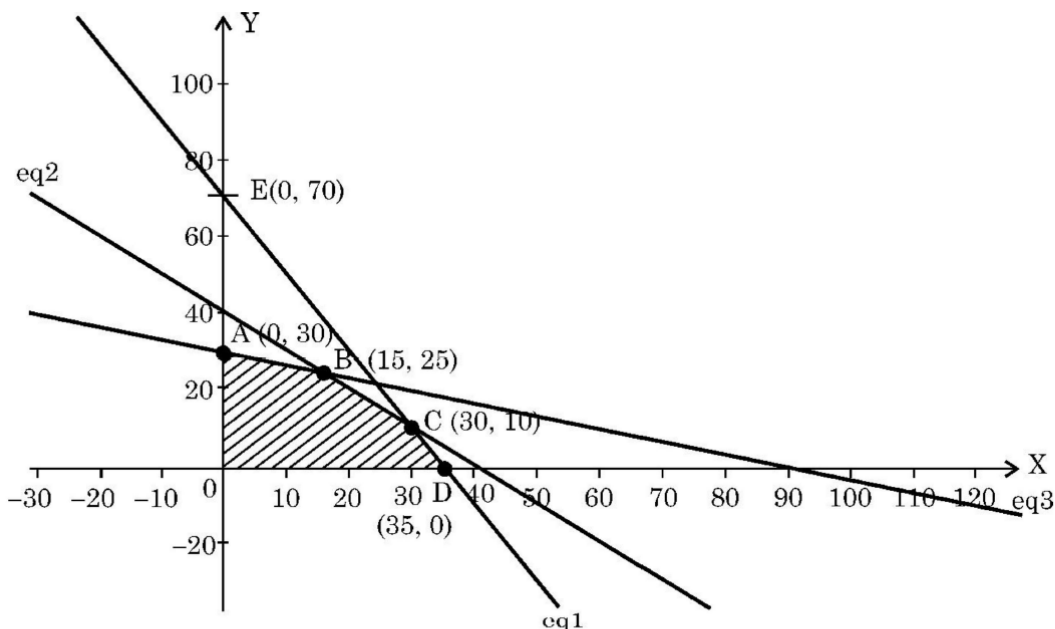


Reason (R) : The optimal solution of a LPP having bounded feasible region must occur at corner points.

27.	Solve the following linear programming problem graphically : Maximise $Z = 2x + 3y$ subject to the constraints : $x + y \leq 6$ $x \geq 2$ $y \leq 3$ $x, y \geq 0$
28.	Solve the following L.P.P. graphically : Maximise $Z = 60x + 40y$ Subject to $x + 2y \leq 12$ $2x + y \leq 12$ $4x + 5y \geq 20$ $x, y \geq 0$
29.	The corner points of the feasible region in graphical representation of a L.P.P. are (2, 72), (15, 20) and (40, 15). If $Z = 18x + 9y$ be the objective function, then (A) Z is maximum at (2, 72), minimum at (15, 20) (B) Z is maximum at (15, 20) minimum at (40, 15) (C) Z is maximum at (40, 15), minimum at (15, 20) (D) Z is maximum at (40, 15), minimum at (2, 72)
30.	If the feasible region of a linear programming problem with objective function $Z = ax + by$, is bounded, then which of the following is correct ? (A) It will only have a maximum value. (B) It will only have a minimum value. (C) It will have both maximum and minimum values. (D) It will have neither maximum nor minimum value.

31.	Solve the following linear programming problem graphically : Maximise $Z = x + 2y$ Subject to the constraints : $x - y \geq 0$ $x - 2y \geq -2$ $x \geq 0, y \geq 0$
32.	A factory produces two products X and Y. The profit earned by selling X and Y is represented by the objective function $Z = 5x + 7y$, where x and y are the number of units of X and Y respectively sold. Which of the following statement is correct ? (A) The objective function maximizes the difference of the profit earned from products X and Y. (B) The objective function measures the total production of products X and Y. (C) The objective function maximizes the combined profit earned from selling X and Y. (D) The objective function ensures the company produces more of product X than product Y.
33.	Assertion (A) : Every point of the feasible region of a Linear Programming Problem is an optimal solution. Reason (R) : The optimal solution for a Linear Programming Problem exists only at one or more corner point(s) of the feasible region.

34.



The feasible region along with corner points for a linear programming problem are shown in the graph. Write all the constraints for the given linear programming problem.

CASE STUDY

35.

The month of September is celebrated as the Rashtriya Poshan Maah across the country. Following a healthy and well-balanced diet is crucial in order to supply the body with the proper nutrients it needs. A balanced diet also keeps us mentally fit and promotes improved level of energy.

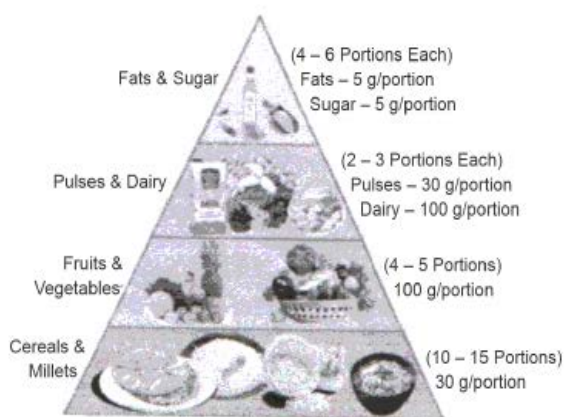


Figure-1

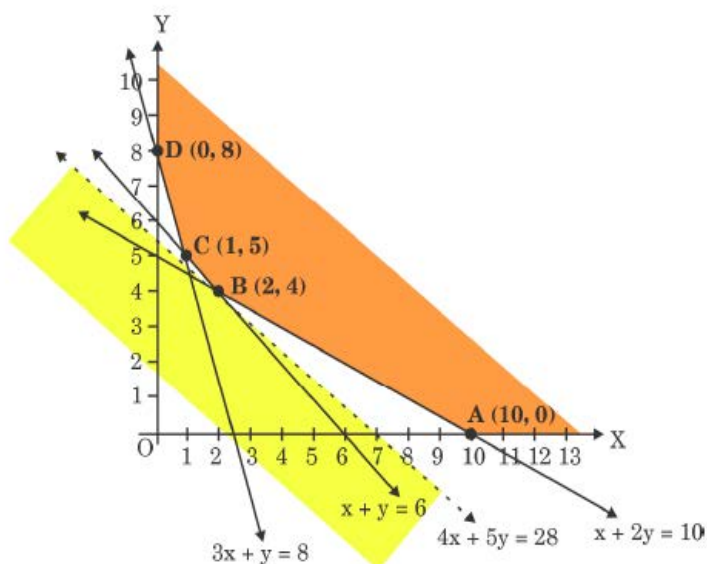
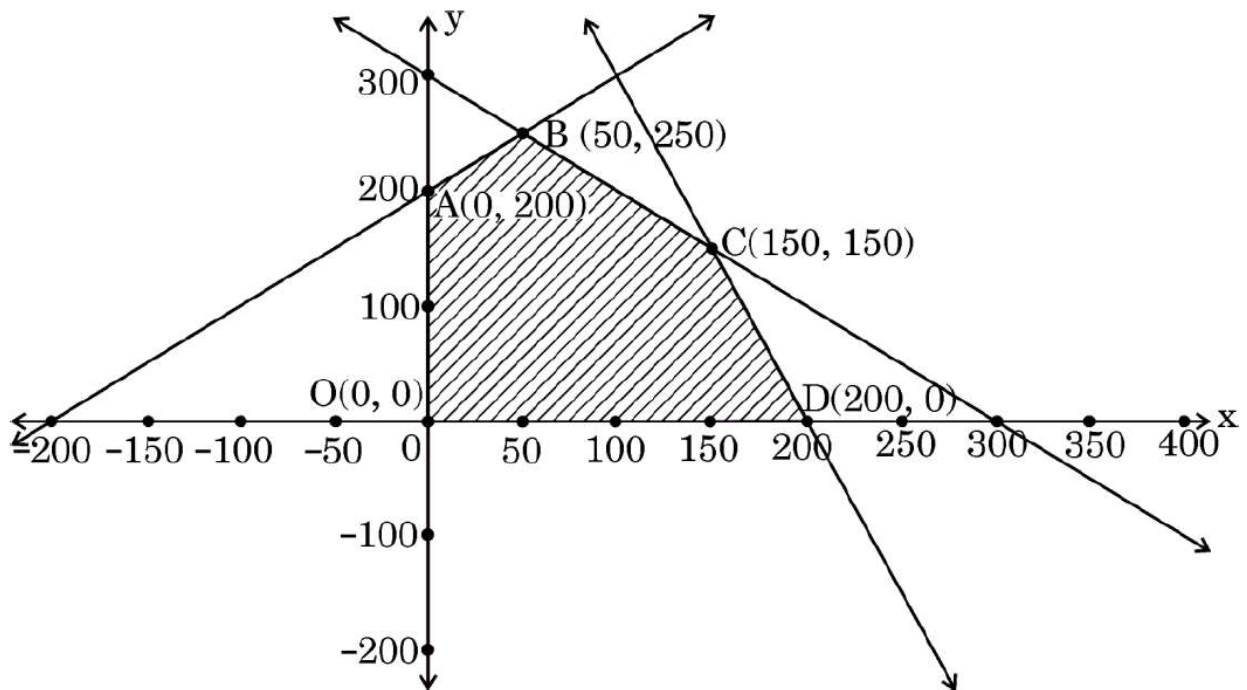


Figure-2

	A dietician wishes to minimize the cost of a diet involving two types of foods, food X (x kg) and food Y (y kg) which are available at the rate of ₹ 16/kg and ₹ 20/kg respectively. The feasible region satisfying the constraints is shown in Figure-2. On the basis of the above information, answer the following questions : (i) Identify and write all the constraints which determine the given feasible region in Figure-2. (ii) If the objective is to minimize cost $Z = 16x + 20y$, find the values of x and y at which cost is minimum. Also, find minimum cost assuming that minimum cost is possible for the given unbounded region.
36.	The corner points of the feasible region of a Linear Programming Problem are (0, 2), (3, 0), (6, 0), (6, 8) and (0, 5). If $Z = ax + by$; ($a, b > 0$) be the objective function, and maximum value of Z is obtained at (0, 2) and (3, 0), then the relation between a and b is : (A) $a = b$ (B) $a = 3b$ (C) $b = 6a$ (D) $3a = 2b$
37.	<i>Assertion (A) :</i> In a Linear Programming Problem, if the feasible region is empty, then the Linear Programming Problem has no solution. <i>Reason (R) :</i> A feasible region is defined as the region that satisfies all the constraints.

38.



For the given graph of a Linear Programming Problem, write all the constraints satisfying the given feasible region.

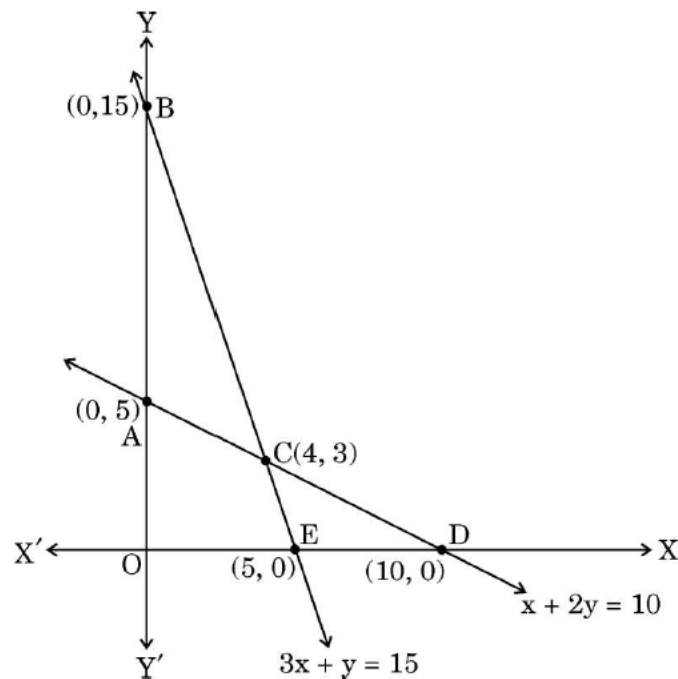
39.

For a Linear Programming Problem (LPP), the given objective function $Z = 3x + 2y$ is subject to constraints :

$$x + 2y \leq 10$$

$$3x + y \leq 15$$

$$x, y \geq 0$$



The correct feasible region is :

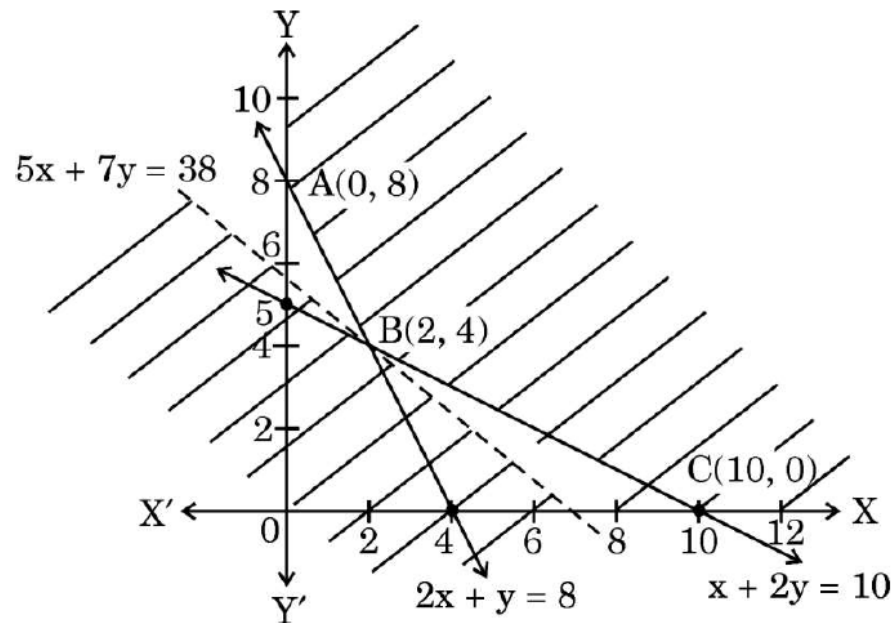
(A) ABC

(B) AOEC

(C) CED

(D) Open unbounded region BCD

- 40.** Assertion (A) : The shaded portion of the graph represents the feasible region for the given Linear Programming Problem (LPP).



$$\text{Min } Z = 50x + 70y$$

subject to constraints

$$2x + y \geq 8, \quad x + 2y \geq 10, \quad x, y \geq 0$$

$Z = 50x + 70y$ has a minimum value = 380 at B(2, 4).

Reason (R) : The region representing $50x + 70y < 380$ does not have any point common with the feasible region.

- 41.** In the Linear Programming Problem (LPP), find the point/points giving maximum value for $Z = 5x + 10y$
subject to constraints

$$x + 2y \leq 120$$

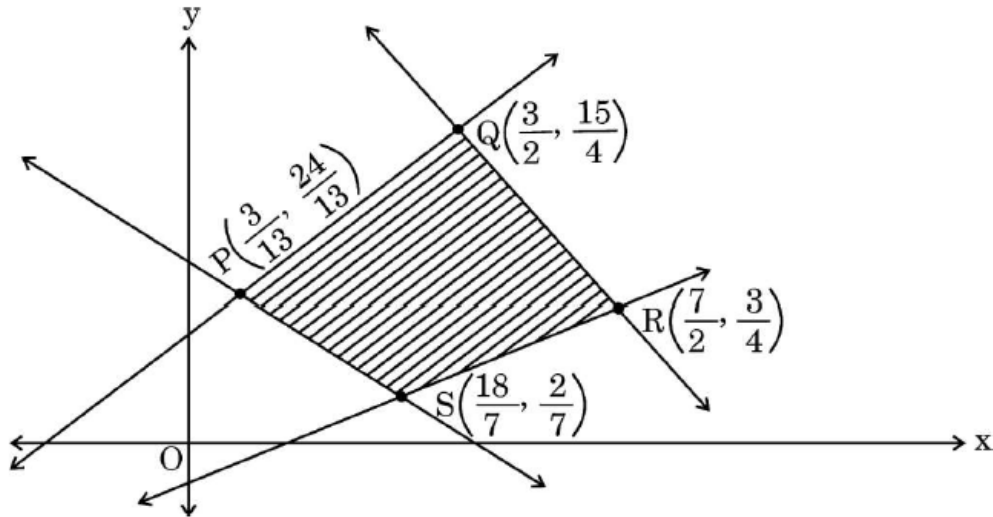
$$x + y \geq 60$$

$$x - 2y \geq 0$$

$$x, y \geq 0$$

42.

For a Linear Programming Problem (LPP), the given objective function is $Z = x + 2y$. The feasible region PQRS determined by the set of constraints is shown as a shaded region in the graph.



(Note : The figure is not to scale)

$$P \equiv \left(\frac{3}{13}, \frac{24}{13}\right), Q \equiv \left(\frac{3}{2}, \frac{15}{4}\right), R \equiv \left(\frac{7}{2}, \frac{3}{4}\right), S \equiv \left(\frac{18}{7}, \frac{2}{7}\right)$$

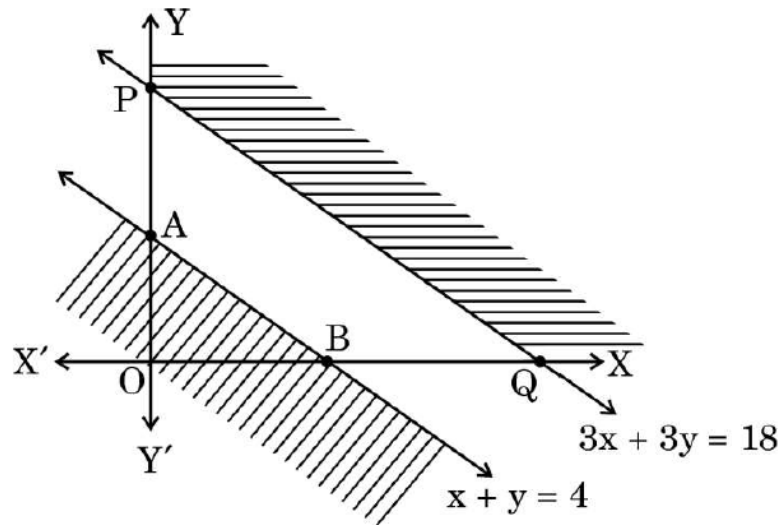
Which of the following statements is correct ?

- (A) Z is minimum at $S\left(\frac{18}{7}, \frac{2}{7}\right)$
- (B) Z is maximum at $R\left(\frac{7}{2}, \frac{3}{4}\right)$
- (C) (Value of Z at P) > (Value of Z at Q)
- (D) (Value of Z at Q) < (Value of Z at R)

43. In a Linear Programming Problem (LPP), the objective function $Z = 2x + 5y$ is to be maximised under the following constraints :

$$x + y \leq 4, \quad 3x + 3y \geq 18, \quad x, y \geq 0$$

Study the graph and select the correct option.



(Note : The figure is not to scale)

The solution of the given LPP :

- (A) lies in the shaded unbounded region.
 (B) lies in $\triangle AOB$.
 (C) does not exist.
 (D) lies in the combined region of $\triangle AOB$ and unbounded shaded region.

44. Consider the Linear Programming Problem, where the objective function $Z = (x + 4y)$ needs to be minimized subject to constraints

$$2x + y \geq 1000$$

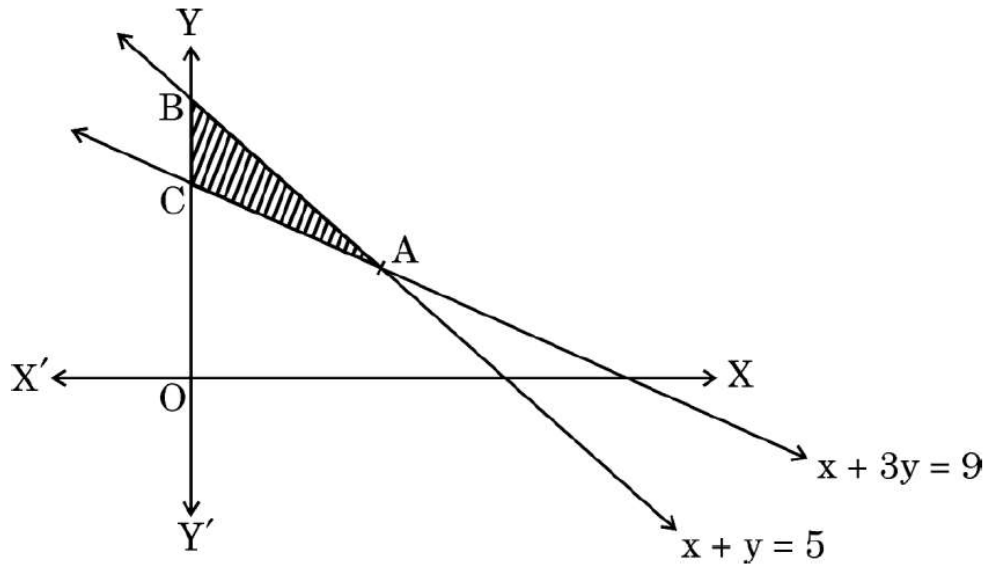
$$x + 2y \geq 800$$

$$\mathbf{x}, \mathbf{y} \geq \mathbf{0}.$$

Draw a neat graph of the feasible region and find the minimum value of Z.

45.

For a Linear Programming Problem, find $\min Z = 5x + 3y$ (where Z is the objective function) for the feasible region shaded in the given figure.



(Note : The figure is not to scale)

46.

In the Linear Programming Problem for objective function $Z = 18x + 10y$ subject to constraints

$$4x + y \geq 20$$

$$2x + 3y \geq 30$$

$$x, y \geq 0$$

find the minimum value of Z .